



Flying Liferaft

Propulsion Architecture, Engine and its Systems



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Abstract

To save lives at sea, a small but fast UAV with a response time faster than a helicopter or coast guard vessel must be deployed. After determining the baseline configuration and mission profile, it was determined that the parallel hybrid-electric propulsion system was the best propulsion method for this UAV because it contributes the least to total weight and has a simple structure. During the analysis of the propulsion methods, the two-stroke piston engine was also chosen because of its compactness and high power-to-weight ratio. Furthermore, the components of the engine control system such as clutches, starter motor and throttle servo were designed and chosen to allow for a robust autonomous operation. Despite some difficulties in estimating the fuel mass and fuel consumption for the chosen engine due to a lack of experimental data, the fuel was estimated using a comparable size engine with the same displacement stroke, which provides an adequate estimate of the fuel consumption for this design phase.

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1 Introduction

1.1 Motivation

Due to civil wars and economic downturns in some countries, many migrants and refugees from North Africa, the Middle East, and South Asia decided to travel to Europe via the Mediterranean Sea in search of better jobs and opportunities[1]. There were approximately 20,000 people who lost their lives in the Mediterranean Sea between 2014 and 2019 which makes it the World's deadliest migration route[2]. To increase the possibilities of the migrants surviving from distresses at sea, the challenges were to be able to detect the migrant's position while also getting there as fast as possible and delivering floatable equipment such as life rafts, preventing them from drowning and reducing the possibility of suffering from hypothermia in cold water. Furthermore, in order to address the aforementioned challenges while minimising response time, a compact vehicle capable of carrying payloads, cruising at high speeds, and requiring only a small setup time must be deployed.

Unmanned aerial vehicles (UAV) are gaining popularity in a variety of applications, including search and rescue (SAR) missions. Although technological advancements have resulted in more powerful batteries and highly integrated electronics and subsystems that improve small UAV performance, there are still major concerns with its significant constraint on short flight time[3]. Therefore, the aim of this group design project is to develop a prototype of a compact high-speed UAV with a longer endurance, thereby shortening the time to target and increasing the survivability of the people in distress further away from the coast. With a more sophisticated mission profile, a single energy system may not be able to achieve all of the desired requirements; thus, the integration of hybrid systems that combine the benefits of various energy sources will be explored.

1.2 Mission profile

The mission profile is divided into five stages. The vertical take-off and transition into forward flight make up the first stage. The second stage is cruising toward the people in distress at the sea. The UAV will then descend for loitering and hovering to identify people in distress and deliver the life raft in the third stage. The UAV is expected to transit to a forward flight to return to base after delivering the life raft.

1.3 Objectives

The task of the FLR04 sub-team was to design the propulsion and UAV system required for the settled configuration to complete the defined mission profile successfully and efficiently. The task involves the propeller and power plant design and sizing, power plant performance and estimation, subsystem selection and design, and weight and balance estimation.

This report is part of the propulsion and system sub-team, which focuses on power plant configuration and component selection in order to reduce weight and increase efficiency. It also briefly covers power plant integration to demonstrate how the selected components will be linked together and function as a whole system to meet the propulsion requirements.

2 Propulsion System Configuration

Considering the mission profile, four VTOL propellers and one pusher propeller were proposed for the baseline configuration[4]. During the VTOL stage, the drone takes off vertically, producing vertical thrust with the four VTOL propellers while the pusher propeller is inoperative, causing the UAV to behave like a quadrotor. When the UAV reaches the desired altitude, it enters forward flight by activating the pusher propeller to gain speed. While the pusher propeller continues to produce thrust, the four VTOL propellers are turned off once the stall speed is reached, causing the UAV to behave similarly to a conventional fixed-wing aircraft. According to the literature[5], most propulsion systems for UAVs with five propellers fall into four different configurations: electric, turboelectric, series, and parallel hybrid-electric as illustrated in Figure 1.

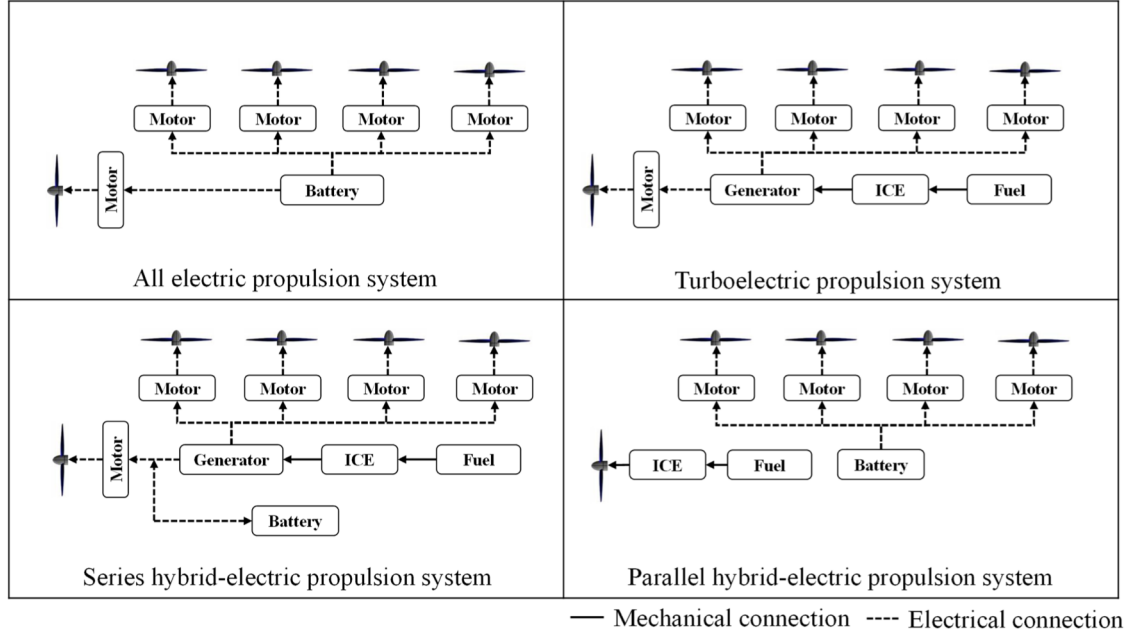


Figure 1: Schematic diagram of four propulsion systems [5]

After the propeller design was completed, it was determined that each VTOL motor required 2.8 kW for take-off and the pusher propeller required a maximum power of 7.8 kW[4]. This value will be used to determine the best configuration for this UAV based on the weight contribution of each design.

2.1 All-electric propulsion system

All-electric propulsion systems rely solely on the battery to power both the propellers and the onboard subsystems. When electricity is used as the source of energy, the efficiency of the system improves when compared to a traditional system that uses a combustion engine. This is due to the use of electric motors which can be up to 95% efficient compared to 18%-23% for traditional combustion engines[6]. Moreover, it is more reliable due to a lower risk of mechanical failure and does not produce as much heat as a combustion engine, which means that it does not necessitate as much cooling as a combustion engine. Additionally, it is quieter and produces less vibration, allowing the UAV to deliver the payload with greater precision and control.

Batteries, on the other hand, store less energy by mass and volume than fuels, so bulky and heavy batteries must be used to achieve the same endurance as fuel. Moreover, the time it takes to recharge is larger than the time it takes to refill the fuel, limiting the amount of time taken before it can be re-launched. This means that the operator must keep a spare battery on hand in order to re-launch promptly after one flight. Furthermore, battery health must be maintained, especially when the UAV is not in use for an extended period of time, as battery health can have a major impact on endurance, raising maintenance costs.

2.2 Turboelectric propulsion system

Turboelectric propulsion system involves a generator to convert the mechanical energy from the internal combustion engine to electrical energy to power the electric motors. This excludes the need of heavy and complex gearbox in the system. Furthermore, it eliminates the need for large batteries to power all of the motors, potentially reducing the weight of the propulsion system. This allows the UAV to have a longer endurance than if it relied solely on battery power because fuel has a higher energy density than battery power. Additionally, the difference in mass between the two configurations can be replaced with fuel to increase the endurance.

On the other hand, the downside of this configuration is that there will be losses when converting from one form of energy to the another, hence a larger engine capable of producing more power

than what the generator required must be selected to compensate for those losses. This reduces fuel efficiency because more fuel must be burned due to the larger engine required. Furthermore, the generator must be capable of producing enough power to power all four VTOL propellers as well as the pusher propeller, resulting in a heavier generator to satisfy the power requirements.

2.3 Hybrid propulsion system

A hybrid propulsion system is one that is powered by a combination of two or more energy sources. It combines the advantages of various energy sources, the most common of which is battery and fuel. The hybrid propulsion system can be connected in two ways: series or parallel.

2.3.1 Series hybrid-electric propulsion system

As shown in Figure 1, the proposed series hybrid-electric propulsion system modifies the turbo-electric propulsion system by adding a battery between the motor and the generator. Since the pusher propeller will be the most active while performing the mission profile, this configuration may increase fuel efficiency because the ICE can reduce power while the battery is also operative. Additionally, the battery can also be charged during flight, eliminating the need to charge the battery separately via charging port. Moreover, this is advantageous because the battery can also power the starting motor, allowing the engine to be started remotely. Lastly, because of the combination of power from the battery and the generator, the battery does not have to be very large, potentially reducing the overall weight of the propulsion systems.

However, the chosen generator must be large enough to provide enough power for the four motors to take off. Furthermore, because the VTOL propeller is powered by the engine, the engine must be started prior to takeoff. Moreover, because losses will occur during the mechanical energy to electrical energy conversion, the engine must be able to provide more power than the generator, necessitating a larger engine and an increase in weight. Additionally, this configuration includes the more components than the parallel configuration as shown in Figure 1, so it is expected to be heavier and more complex, resulting in higher costs when compared to the parallel configuration.

2.3.2 Parallel hybrid-electric propulsion system

The parallel hybrid-electric propulsion system depicted in Figure 1 is composed of two distinct systems: one powered by battery and another one by fuel. This means that during cruise phase, the motors can be turned off, which results in a smaller battery compared to the all electric configuration as the VTOL will only be used in takeoff, landing and short hover to deliver payload only. Another advantage is that in the event that one of the systems fails, the UAV can still switch to the other operable system and prepare for landing. In the event that the battery fails, the UAV can rely on the pusher propeller to slow down to stall speed before turning it off and performing an emergency landing with the landing buoys carried on board.

However, the disadvantage of this configuration compared to the series configuration is that the fuel consumption will be high as the engine is the only source of power to produce power for the pusher propeller. Moreover, the battery must be sized to power the four VTOL propellers, resulting in a heavier battery.

2.4 Comparison of each configuration and selection

Table 1: Comparison of the total power plant mass for each configuration

Configuration	Total Mass(kg)
All electric	48.880
Turboelectric	20.850
Series hybrid-electric	24.516
Parallel hybrid-electric	12.684

Table 1 compares the masses of the power plant, including the battery, generator, engine, and the required fuel mass to complete the mission profile for each configuration. This comparison assumes that the fuel mass required is the same for every configuration as there is a limited data on fuel consumption for each engines. A break down of masses can be found in Appendix A.

When compared to other configurations, the all-electric propulsion system contributes the most weight due to the required flight time of one hour. Because batteries can only store a small amount of energy when compared to fuel, this configuration requires a large battery to produce the same amount of power as fuel, which is why it was the heaviest and thus eliminated. Furthermore, because the turboelectric propulsion system relies on the combustion engine and generator to generate power for all of the motors, the engine must be large and capable of producing more power to compensate for the energy loss during the conversion process. Additionally, because the four VTOL propellers will only be used during takeoff, landing, and short hover, it is not worth adding extra weight to the engine just to have enough power for these motors for such a short period of time, so this configuration was not chosen. Similarly, the proposed series hybrid-electric system is configured similarly to the turboelectric system, but with the addition of a battery connected to the pusher propeller. Although this may reduce the fuel consumption as the pusher propeller will be powered by two sources of energy, but the weight of the engine will need to remain the same as still needs to be able to power the four VTOL propellers, and it also need the extra power to recharge the battery.

Finally, the chosen configuration for this UAV was the parallel hybrid-electric system because it has the lowest weight among the proposed configurations. This is due to the engine only having to power the pusher propeller and the battery only having to power the four VTOL motors for a short period of time. This configuration is the best compromise between engine and battery use, resulting in the lightest propulsion system to perform the defined mission profile.

3 Charging System Selection

Following the selection of the configuration, the method of charging the battery was considered. The integration of an alternator and the attachment of solar panels to the wing are two methods that have been considered. This section will explain why these charging methods were not implemented in this UAV prototype. Finally, it was decided that the UAV will adopt a traditional charging port to charge the battery. Table 2 shows the required specifications for charging the selected bank battery.

Table 2: Battery specification taken from N. Melo Fernandez [7]

Battery	18650-3030A
Voltage (V)	46.8
Battery Capacity (Ah)	27
Charging current range (A)	1.3-39

3.1 Generator

A generator can be connected to the engine via pulley and belt to generate electricity from the mechanical energy provided by the combustion engine, which can then be used to charge the battery. The generator has the advantage of being able to charge the battery while the UAV is in operation, as well as being small in size. The generators on the market that can potentially be used for the charging system are listed in Table 3.

To match the voltage required to charge the battery, the current displayed on Table 3 is the output current after using a step up transformer for generators with voltages less than 46.8V. Additionally, the current for 48V generator is limited to 39A, which is the maximum charging current of the battery, in order to preserve its health. The extra fuel mass required is an approximation based on the specific fuel consumption of a two-stroke DA120 engine[8], with the engine running at idle speed to charge the battery on the ground.

According to Table 3, the 48V generator must be used to charge to full within flight, but the added mass to the system is significant, so it was not chosen. Next, while both the 24V and 12V generators have comparable mass, the charging time for the 24V generator is shorter, so the 12V generator was eliminated. Furthermore, the estimated flight time to complete the full range mission profile is one hour, implying that the battery cannot be charged to full during operation and must be charged separately on the ground as the charging time for 24V generator is estimated to be 4.32 hours. To charge the battery on the ground, a fuel mass of 3.82 kg is required, making it impractical and a waste of resources to simply burn fuel to charge the battery only. Moreover, when the UAV does not operate frequently, 3.82 kg of fuel must be burned to preserve the battery's health. As a result, the idea of incorporating a generator into the propulsion system for charging was eliminated.

Table 3: Specification of generators

Generator	96 Series Alternator	ATO-PMG-NE200WS	Car Builder Solution
Voltage (V)	48	24	12
Current (A)	39	5	4
Charging time (hr)	0.55	4.32	6.75
Mass (kg)	9.0	3.4	3.0
Extra fuel mass required (kg)	0.49	3.82	5.97

3.2 Solar panel

Adopting solar panels onto the UAV wing could be used to recharge the battery during flight, eliminating the need for overhead equipment such as a spare battery and possibly increasing the UAV's endurance. However, when attaching the solar panel to the wings, some difficulties arise because the photovoltaic cells must be protected from the wind and must be able to flex when air passes through the wing, but solar cells are typically made of glass and are not very flexible. Moreover, solar panels are only 20% efficient in converting sunlight to electricity[9]. This means that many panels will be needed to have a reasonable recharge time, however, due to the compactness requirements, the wing will not have enough area to support all the panels required to recharge the battery. Another issue with relying on solar panels is that they only work during the day, and an open storage is required to recharge and maintain the battery health when it is not frequently used.

Table 4: The specification of 3 monocrystalline solar panels connected in series to satisfy the battery charging requirements [10]

Optimum Operating Voltage (V)	56.70
Optimum Operating Current (A)	5.29
Mass added (kg)	5.70
Charging time (hr)	4.08
Dimension per panel (cm)	121.92x54.90x0.20

Table 4 shows that the additional weight is significant, while the required charging time when the battery is at 20% capacity is 4 hours. Furthermore, this calculation was overestimated because it does not account for gloomy weather where sunlight availability is limited, which may increase charging time. Finally, the wing does not have enough space to carry all of the three panels due to compactness requirements, as a result, this method was eliminated.

4 Internal Combustion Engine

4.1 Types of Engines

The types of engines used in UAVs are determined by the task and the environment in which they operate. In most UAV applications, engines such as turboprops, turbopumps, Wankel, and pistons are used.

Table 5: Example of UAVs with different types of engines

UAV	Engine Type	MTOW (kg)
Neptune	Piston	36
Lihniniya MK I	Two-stroke piston	52
RQ-7A Shadow 200	Wankel	148
Lockheed Martin RQ-3 Dark Star	Turbofan	3860
MQ-9 Reaper	Turboprop	4760

Based on the examples in Table 5, UAVs with similar MTOW ranges to this project’s prototype use piston engines; hence, piston engines were chosen to be investigated further for the internal combustion engine. Furthermore, because of a better knowledge of the concepts underlying the Wankel rotary engine and its recent development[11], it was also taken into consideration as it may provide advantages over the piston engines.

4.2 Piston Engine

4.2.1 Two-stroke

The two-stroke piston engine is a reciprocating engine that uses two-time movement to produce a power stroke. During the intake, the fuel-air mix enters the chamber and is compressed by the upward motion of the piston. The compression causes the mixture’s pressure and temperature to rise slightly. Next, when the fuel encounters the spark plug, combustion occurs as a result of the spark plug ignition. After the combustion, the thermal energy in the chamber causes the piston to move downwards, increasing the volume of the chamber which is known as the power stroke[12]. The exhaust then leaves the chamber causing a decrease in pressure and the cycle is repeated. Since two-stroke engines do not have valves, they are simpler and 30% lighter than four-stroke engines[13]. Furthermore, two-stroke engines fire once per revolution, resulting in a high power-to-weight ratio when compared to four-stroke engines of comparable size. The two-stroke piston engine also has the advantage of already having the lubricant mixed with the fuel, allowing it to be placed in any orientation[13].

On the other hand, because it fires once every revolution, it has low fuel efficiency. It also has a high vibration and noisy operation as well as incomplete combustion which contributes significantly to pollution. Additionally, it is more fragile and undergoes more wear during operations, resulting in a shorter life span than four-stroke engines[13].

4.2.2 Four-stroke

A four-stroke piston engine completes one complete cycle by using four strokes: intake, compression, power, and exhaust. The steps for compression and power stroke are similar to those of two-stroke engines, with the exception that four-stroke engines have an intake stroke in which the piston moves down, creating a vacuum that draws fuel-air mix into the chamber through an open intake valve[14]. It also includes an exhaust stroke in which the piston moves upward, and the intake valve closes, allowing the exhaust valve to open[14]. This stroke forces the exhaust gas out of the engine and the process is then repeated. Because fuel is consumed only once every four strokes, it has a higher fuel efficiency than two-stroke engines. Furthermore, it emits less pollution because power is generated only once every four strokes and no oil or lubricant is added to the fuel[15].

On the contrary, the power-to-weight ratio of four-stroke engines is lower than that of two-stroke engines, requiring a heavier engine to produce the same amount of power. Furthermore, because it involves complicated mechanisms such as complex valves, the maintenance costs are high in order to troubleshoot it[15].

4.3 Wankel Rotary Engine

The Wankel engine employs a rotor design that directly transforms the pressure energy of gases into rotary motion as it completes a full cycle [16]. The new mixture reaches the first chamber

and is ready for combustion when the rotor tip passes the intake valve. As the rotor revolves, the fresh fuel-air mixture is compressed by the engine geometry by the time it reaches the spark plug. When the spark plug ignites, the highly compressed fuel-air mixture explodes, and the expansion gas forces the rotor forward toward the exhaust pipe, allowing hot air to leave[16]. Additionally, this movement occurs synchronously during the four stages, resulting in a more reliable engine when compared to piston engines. Furthermore, it has fewer moving parts, which simplified engine construction, and it weighs less than a four-stroke piston engine delivering the same power output, resulting in a better power-to-weight ratio[16].

However, the sealing problem arises in Wankel engines due to the fact that each chamber operates at different temperatures. Moreover, the combustion is slow because of the lengthy, thin moving portions, which increases the potential of incomplete combustion, in which fuel is discharged without even being combusted[16]. As a result, Wankel engines cannot burn fuel as cleanly or efficiently as piston engines due to the unburnt fuel in the exhaust.

4.4 Shortlisted Engine

Table 6 summarises the shortlisted engines based on the power required for the pusher propeller of 7.8kW obtained from the propeller design. The specific fuel consumption (SFC) shown on Table 6 is the optimum SFC of the engine. The manufacturer did not provide the SFC of the DLE120 engine, so the optimum SFC data was obtained from a DA120 engine data sheet[8], which has a similar stroke displacement. Furthermore, because Wankel rotary engine does not provide its SFC, it was not included in Table 6.

Table 6: Shortlisted engine

Engine	Engine Type	Mass (kg)	Power (kW)	SFC (g/kWh)
DLE120	Two-stroke piston	2.9	8.95	550
DF140LC	Four-stroke piston	6.3	8.61	330
80S 15BHP (AIE)	Wankel rotary	5.0	11.20	-

Firstly, the 80S Wankel rotary engine was eliminated because it requires a liquid cooling system, which requires more mass to be added to the system. Incomplete combustion also results in low fuel efficiency and high emissions.

The DLE120 two-stroke piston engine was found to be the most suitable engine for the UAV prototype after comparing the pros and cons between two-stroke and four-stroke piston engine. Table 6 clearly shows that the weight of the chosen engine is the lightest among the shortlisted engines providing a comparable power output. This also implies that by using this engine, there are potential spaces left for batteries to improve hovering endurance. Moreover, because the purpose of the UAV is to assist people in distress at sea, it is not expected to operate on a daily basis like commercial aircraft, so the engine's lifespan will not be significantly affected, as is typical of a two-stroke engine. Furthermore, because the estimated flight time is less than one hour, pollution is expected to have a minimal impact on the environment. The engine's accessories, such as mufflers, can be used to reduce noise and vibration. Additionally, DLE120 includes an air-cooled system, which allows heat to be dispersed from the engine fin as long as there is enough air ventilation passing through the engine, reducing the need for coolant and thereby minimising extra weight added.

By comparing the price of the shortlisted engine, the two-stroke piston engine is also the cheapest due to its simple structure. Although the two-stroke piston engine has a lower fuel efficiency than the four-stroke piston engine, the combined fuel and engine weight of the two-stroke piston is 1.684 kg less than the four-stroke engine.

The spool up time of the DLE120 engine was investigated further after the propeller characteristics were finalised. It determines whether there is enough excess power to cause the propeller to accelerate fast enough to be controllable. It was found that with the designed propeller, the spool up time of the DLE120 engine was within a reasonable range, allowing the engine to be controllable[17]. As a result, this confirms that the engine chosen is appropriate for this UAV.

5 Propulsion System Components Selection

5.1 Electric motors

Four propellers will be powered solely by the battery, necessitating the use of electric motors. It is critical to choose an appropriate motor because the current drawn by motors and voltage constraints have a significant impact on battery sizing. After the rotor's power requirements for VTOL and diameter have been confirmed, the electric motors were chosen. The power required to take off for each motor is 2.8kW, and the diameter is 0.6 m, both of which were determined from the propeller design calculations [4]. Since there is a limited amount of experimental data available for motors with different propeller sizes on the market, the shortlisted motors are those with experimental data available at a similar range of propeller diameter or that are advertised to be able to produce more thrust than the current MTOW. Table 3 presents the shortlisted motors that meet the power and thrust requirements.

Table 7: Experimental data for shortlisted motors

Motor	Propeller diameter (mm)	Max. Power (W)	Voltage (V)	Mass (kg)	RPM	Thrust (g)
U12II KV120	736.66	4526.4	48	0.778	4742	20400
KDE6815XF-205	546.10	4541	46.2-52.2	0.630	8220	14329

Another factor to consider when choosing motors was their operating rpm. This is due to the restriction in the space available to mount the rotors, hence the motors must be capable of reaching a higher rpm in order to reduce the size of the rotors, hence the KDE6815XF-205 motor was chosen.

5.2 Clutch

To ensure the safety of people nearby, the pusher propeller must not spin on the ground during starting and VTOL phases, thus a clutch must be applied to the engine to detach the pusher propeller from the engine. There are ten types of clutches available on the market that operate in various ways and are used for different types of applications[18]. As a result, it must be carefully selected for this UAV application to reduce the complexity of the additional components added. Of ten different types of clutches, there are two clutches that are suitable for the UAV: the electromagnetic clutch and the diaphragm clutch.

5.2.1 Electromagnetic Clutch

An electromagnetic clutch is a friction clutch that engages and disengages the clutch by using magnetic force in the plane of spring force[19]. It is made up of two clutch plates called the rotor and the hub, one of which is connected to the battery and the other to the propeller. There is also a friction plate between them. When a direct current is applied to the plate, it becomes an electromagnet, attracting the other armature and connecting it to the friction plate in the middle, which transmits torque between them [19]. The advantage of using an electromagnetic clutch is that it is initially in the disengaged position, ensuring that the pusher propeller does not spin during the starting process. Moreover, it is very lightweight and the need for extra components is not required, reducing the complexity of the system. It provides fast and smooth operations which is suitable for remote operations. However, the main disadvantage is that it requires battery power to engage the propeller with the engine axle, so the battery must provide enough power for the entire cruise flight.

5.2.2 Diaphragm Clutch

A diaphragm clutch is commonly used in automotive applications where it uses the diaphragm spring to engage and disengage the clutch. The working principle is the opposite of the electromagnetic clutch as it is engaged in the initial position and requires a power input to disengage the clutch. As a result, for safety reasons, the clutch must be double-checked during the starting process to ensure that the power supply is connected to its controller to disengage it. Moreover,

it has a complex mechanism because it requires additional components such as a diaphragm, and solenoid. The solenoid controls the diaphragm mechanism and must be carefully selected because the diaphragm spring is very stiff which requires a lot of power to detach it from the engine axle, adding weight. However, the need for a power supply is minimal as it is always engaged to the engine axle and does not require much maintenance.

5.2.3 Final Selection

After weighing the benefits and drawbacks of the two types of clutches, the electromagnetic clutch was selected due to its compactness and lightweight. Additionally, due to the fuselage space constraints, using an electromagnetic clutch is more beneficial than using a diaphragm clutch because no other additional components are required to operate the clutch. Moreover, having a clutch disengaged in its initial state is safer in the event of a battery failure.

Table 8: Specification of the selected clutch

Type of clutch	Voltage (V)	Mass (kg)	Power required (W)
Electromagnetic	24	0.46	11

5.3 Engine Starter

The method of starting the engine was also taken into consideration in order to ensure an efficient and safe start. There are four common ways to start the engine: hand-propping, electric drill start, pull start, or with the starter motor integrated into the engine.

5.3.1 Hand-propping and electric drill starter

Most of the small UAVs on the market do not have a starting motor, therefore it must be started manually. This is done by spinning the propeller by hand to warm up the engine and manually starting the compression and ignition process. Hand-propping reduces the number of extra components required for an electric starter, resulting in weight savings. However, this procedure is dangerous and must be carried out by trained personnel. The consequences of neglecting to mitigate the hazards associated with hand propping can result in serious injury or death.

Similarly, an electric drill start is also used to start the engines manually. Although compared to the hand-propping method it does not require as much human power to start the engine. However, it does not satisfy the safety requirements as the propeller must spin in order for the engine to start, so the level of danger is comparable. Furthermore, using an electric drill start necessitates the use of additional overhead equipment which does not satisfy the aim of this project.

5.3.2 Pull-start

A pull start is commonly used to start small internal combustion engines on small appliances such as lawn mowers, small outboard motors, and small go-karts. The starter typically includes a rope with a grip or a handle at one end that is wrapped around the engine's crankshaft. Pulling the rope causes the rope to uncoil, tension the spring, engage the clutch, and induce a spinning motion that cranks the engine, allowing it to start before reaching the end of the pull stroke[20]. When the engine is started, the clutch disengages and the rope is retracted by the spring-operated reel, preparing it for the next start operation[20]. This method starts the engine while the clutch on the engine axle is disengaged, so the pusher propeller does not spin. Moreover, the added weight is also minimal when compared to electric starter motors, and it is also safer to use than the hand-propping method. Furthermore, all of the components for starting will be integrated into the UAV, eliminating the need for any additional overhead equipment. However, because the engine would need to be modified to accommodate the pull-start equipment, the weight difference between the electric motor and the pull-start could be negligible.

5.3.3 Electric starter motor

The starter motor is an electric motor that cranks the engine to start. It uses the same principle as the other two methods, but it uses electricity to spin the engine's crankshaft, hence no human power is required in the starting process. One obvious advantage is that it allows the engine to be started during the transition phase rather than at the start of the VTOL, saving fuel weight. Additionally, it is also very user-friendly and does not necessitate the use of highly trained professionals. However, to ensure that the engine has been started and is operational, it is best to start the engine on the ground, as batteries are very common to fail[21]. Furthermore, because the starter motor requires a high current, the battery must have sufficient power, adding weight to the system. Moreover, the weight of the starter motor and its associated components is larger compared to a pull start and increases the complexity of the system.

5.3.4 Final Selection

Since there is a requirement that the pusher propeller must be disengaged on the ground, during the start and end of the VTOL phase, the hand-propping and electric drill starters were initially eliminated. By comparing the pull start and the electric starter motors, it was determined that the electric start motor will be adopted for this UAV. This is due to the flexibility of having the option to start during the flight which saves fuel weight. Moreover, it is user-friendly which means that it is simple to operate and does not require human power to start the engine, allowing the user to start the UAV from a safe distance. The increase in battery weight is also minimal because the motor will only run for less than 10 seconds during the starting process. Finally, when compared to other methods, using an electric motor allows for the most efficient start. Table 9 compares the mass of each type of starting method, as well as whether additional components are required for each product found on the market.

Table 9: Comparison of each starter methods

Type of starter	Mass added to the UAV (kg)	Extra components required?
Hand-prop	0	No
Electric drill starter	0	Yes
Pull-start	0.4	Yes
DLE120 Electric starter motor	0.778	No

5.4 Engine Throttle Servo

To control the engine's throttle, a rotary throttle servo was chosen. Because it is lightweight and has a quick response time, the throttle can be adjusted with very little delay. To adjust the throttle level, a small rod attached to the servo arm will be connected to the engine's throttle lever, causing it to move when the servo rotates. Furthermore, because it is in rotary motion, it does not require a large space to function. Additionally, it can be easily controlled by connecting its motor to the battery and control system. It also eliminates the need for a heavy gear box because the engine can handle all of the propeller's operable rpm and the throttle can be adjusted directly to control the rpm of the engine and the rotor.

Table 10: Specification of the throttle servo

Actuator	Mass (g)	Input Voltage (V)	Power Required (W)
ANNIMOS 25kg digital servo	80	4.8-6.8	0.034

6 Engine Systems and Integration

6.1 Engine Components Integration

The engine will be connected to the fuel tank, start motor, electromagnetic clutch, throttle servo, and pusher propeller. The starter motor will be mounted onto the engine as shown on Figure 2.

Next, the electromagnetic clutch will be attached to the engine's shaft just after the starter motor frame. Moreover, the propeller will be attached to the clutch via a flange and bolt so that the pusher propeller does not spin when the clutch is disengaged from the engine's shaft. Furthermore, the throttle servo will be placed near the throttle lever on the opposite side of the engine. Lastly, the fuel tank will be connected to the engine via a fuel pipe on the same side as the throttle servo.



Figure 2: An illustration of how the starter motor and its associated components will be connected to the engine[22].

6.2 Fuel Calculations and Fuel Tank Design

The DLE120 engine runs on gasoline with an octane rating of 87-93 or RON 95 in the UK. Due to a lack of experimental data for the DLE120 engine's specific fuel consumption, it was calculated using two different approaches. The first approach was estimating it using an empirical equation provided from Raymer[23] as shown in Equation 1.

$$C_{bhp} = \frac{550\eta_p C}{V} \quad (1)$$

Where V is the velocity in feet per second and η_p was assumed to be 0.7 for a fixed-pitch propeller and C was taken from the graph given in Raymer[23] based on the Mach number.

On the other hand, the second approach was to use an experimental data provided by other manufacturer of a comparable size engine with the same stroke displacement as the DLE120 which is 120cc. The SFC was estimated from the average fuel consumption data of the DA120 engine[8].

Table 11: Comparison of SFC between two approaches

Approach	SFC (kg/kWh)	Fuel Mass (kg)
Raymer	0.245	1.910
DA120	0.590	4.602

The fuel mass was calculated assuming a continuous power of 7.8kW and a flight time of one hour. The SFC values in Table 11 have a large difference, so the DA120 approach was chosen because it should give better estimation of fuel consumption of a two stroke piston engine than Raymer's empirical formula for general piston engines.

Next, the fuel tank was sized based on the fuel mass of 4.602 kg, as shown in Table 12. A 5% safety factor was applied to the required fuel tank volume, which accounts for the volume of expansion[24]. The fuel tank will be made out of high-density polyethylene plastic (HDPE) which is both corrosion resistant and lightweight. Moreover, it is also save in crashes as it is smooth which means that there is no joints that is prone to failure compared to other types of tank. The plastic tank does not produce spark and they can also rebound back to their original shape[25].

Table 12: Fuel tank design

Fuel Mass (kg)	4.602
Gasoline Density (kg/m^3)	740
Fuel Volume (cm^3)	6218.9
Required Fuel Tank Volume (cm^3)	6529.9
Fuel Tank Dimension (cm)	18x19x20
Designed Fuel Tank Volume (cm^3)	6840

After the power requirements for each phase of the mission were determined, the fuel mass required was calculated to determine whether the fuel tank designed was sufficient to complete the mission. The fuel mass required was reduced to 2.9kg, allowing for more space for spare fuel to increase endurance. The break down of fuel consumption can be seen in Appendix B.

6.3 Engine Control

On the UAV, the Engine Control Unit (ECU) will be used to control the engine, accepting input such as engine RPM and throttle position. To achieve the desired velocity, the control system calculates the difference between the measured and desired velocity and then sends a signal to the throttle servo to adjust the throttle settings of the engine. In the event of a malfunction in the equipment, the engine rpm sensor detects the rpm and feeds it back into the ECU to limit the position of the throttle servo, thereby limiting the rpm at maximum and preventing structural failure of the propeller.

7 Evaluation and Next step

The estimated fuel weight is likely to be inaccurate as the data used excluded the load on the engine and was also an approximation based on a comparable-sized engine. Following the analysis and selection of the major components for the engine, the chosen engine could be used to conduct a ground test with the designed propeller to obtain more accurate information about the engine such as fuel consumption and the power it can produce at each rpm. With a more reliable result, the control system sub-team can predict and model a precise performance of the UAV, rather than relying on empirical equations to model the propulsion system. Similarly, because the selected motor was constrained by the VTOL propeller characteristics, it could be tested with the designed propeller to determine the actual thrust produced and power consumption, as this primarily affects battery sizing. Moreover, in the next stage a more detailed design of the integration of each engine components onto the engine can be explored in more depth. Additionally, the engine throttle servo must to be tuned with the engine in order to achieve precise throttle control.

8 Conclusion

To conclude, the objectives of this report were fulfilled using the calculations and information available during this project, since the final weight of the UAV was lower than expected[17], given that the propulsion system contributes to a significant portion of its weight. This is because the analysis performed during the selection process was primarily focused on the compromise between mass, efficiency, and cost of the power plant system. Moreover, while fuel estimation can be inaccurate, the values obtained were adequate for this design phase. Furthermore, all of the major components for the engines were carefully designed and selected in such a way that the engine can be controlled autonomously by the onboard control system.

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Appendix

A Propulsion Configurations

These calculations were based on the power requirements of VTOL and pusher propeller of 11.2kW and 7.8kW respectively.

Table 13: Mass break down of battery only system

VTOL battery (kg)	5.499
Pusher Battery (kg)	43.381
Total (kg)	48.880

Table 14: Mass break down of turboelectric system

Fuel mass (kg)	4.29
15kW generator (kg)	10.00
12.6kW Engine(kg)	6.56
Total (kg)	20.85

Table 15: Mass break down of series hybrid-electric system

Fuel mass (kg)	4.290
Battery Mass(kg)	3.666
15kW generator (kg)	10.000
12.6kW Engine(kg)	6.560
Total (kg)	24.516

Table 16: Mass break down of parallel hybrid-electric system

Fuel mass (kg)	4.290
Battery Mass(kg)	5.499
8.9kW Engine(kg)	2.895
Total (kg)	12.684

B Fuel estimation

The power requirements for cruise 1, cruise 2 and loiter are 7.2kW, 4.7kW, 3.3kW respectively.

Table 17: Break down of fuel consumption at each phase

DA120 SFC (kg/kWhr)	0.59
Cruise 1 (kg/hr)	4.248
Cruise 2 (kg/hr)	2.773
Loiter (kg/hr)	1.947